



Attendance Code:

CMP2804 TEAM SOFTWARE ENGINEERING

Week 9

Attendance Code:



LSEPI*

*Legal, Social Ethical, Professional Issues

What LSEP factors may affect your work as a computing/game dev professional?

- public wellbeing
- computer security
- environmental/sustainability issues
- economic/commercial issues
- accessibility
- IP
- trust in software
- globalisation

Do we as computing and IT professionals....:

Just implement systems

Or ...

Should we consider what they are for/what they do?

*This can be a difficult question to answer....

NEWS

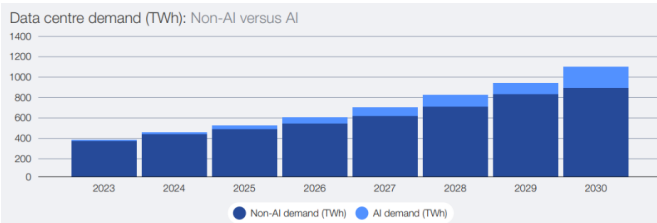
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Technology

Data centres to be expanded across UK as concerns mount



Data centres, like this one Google is building in Hertfordshire, are becoming a more familiar sight across the UK



Note: This is an extrapolated scenario that extends the IEA's forecast from 2023 to 2029 through 2030 using a combination of 2021-2023 historical growth and their proposed growth rate from 2023-2026.
Source: International Energy Agency (IEA); Goldman; Accenture.

NEWS

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Hospital gets low-carbon heating from data centres



Developers 1Energy said Milton Keynes University Hospital had signed on as its "anchor customer"

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Technology

Tiny data centre used to heat public swimming pool

14 March 2023



ard says his scheme can save public swimming

GOV.UK

Home > Environment > Climate change and energy

News story

Thousands of homes to be kept warm by waste heat from computer data centres in UK first

More homes and businesses to benefit from cleaner, low-cost heating - thanks to almost £65 million in government support for 5 green energy projects.

From: Department for Energy Security and Net Zero, The Rt Hon Claire Coutinho MP and Lord Callanan

Published 3 November 2023

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Data centre in the shed reduces energy bills to £40



Therence Bridges says he "can't fault the heating system", which captures heat from more than 500 mini-computers processing data

Ethical frameworks (i.e. the way we think about, reason and understand ethical situations)

- can dictate what is 'right' and 'wrong'

Three types of frameworks are:

Can justify morally questionable actions – if they serve the common good

Utilitarian Ethics

- Questions the impacts (i.e. the consequences) of decisions.
- Assess the benefits/harms before taking a decision
- Eg: *the Online Safety Act 2023. Online platforms protecting users (especially children) from illegal content. Utilitarian might argue that the benefits outweigh the risks.*

Deontological Ethics

- Questions the duties and rights of stakeholders in a decision
- Duty base ethics – follow a set of rules regardless of the consequences.
- Eg: *Apple withholding user data from Government*

Stick to principles – even if they have a negative effect on society ('duty for duties sake')

Virtue Ethics

- Rather than consider the action (deontology), or the consequences (utilitarianism)
 - Focusses on the motivation and virtues of the person
 - Eg: *rather than developing an app to keep engagement, can we develop the app to give the user more self control...?*
-

L Legal

EU General Data Protection Regulations (GDPR)
UK Data Protection Act 2018

UK Equality Act 2010

UK Digital Economy Act 2017

UK Copyright and Rights in Databases Regulations 1997
UK Intellectual Property Act 2014

UK Network and Information Systems Regulations 2018
UK Cyber Security and Resilience Bill (2025 – In Development)

+ many more

S Social

Accessibility

Bias and fairness

Data sets, ML algorithms, etc

Surveillance and tracking

Behaviour and targeted ads, etc



Facebook-Cambridge Analytica scandal

+ Add



Meta investors settle \$8bn lawsuit with Zuckerberg over Facebook privacy

Technology · 17 Jul



Meta settles Cambridge Analytica scandal for \$725m

Technology · 23 Dec 2022



Facebook sued for 'losing control' of users' data

Technology · 9 Feb 2021



Zuckerberg and Hancock meeting details released

UK Politics · 8 Dec 2020



Police officers raise concerns about 'biased' AI data

16 September 2019



GETTY IMAGES

Officers expressed concern biases in the data used to train the tools could direct them to take unfair actions

Attendance Code:

E Ethics

For a computing professional...?

Duty of care

*Could anyone/anything be adversely affected by this software?



Responsibility for consequences

*Could anyone take the deliverable from your project and use them for questionable purposes?

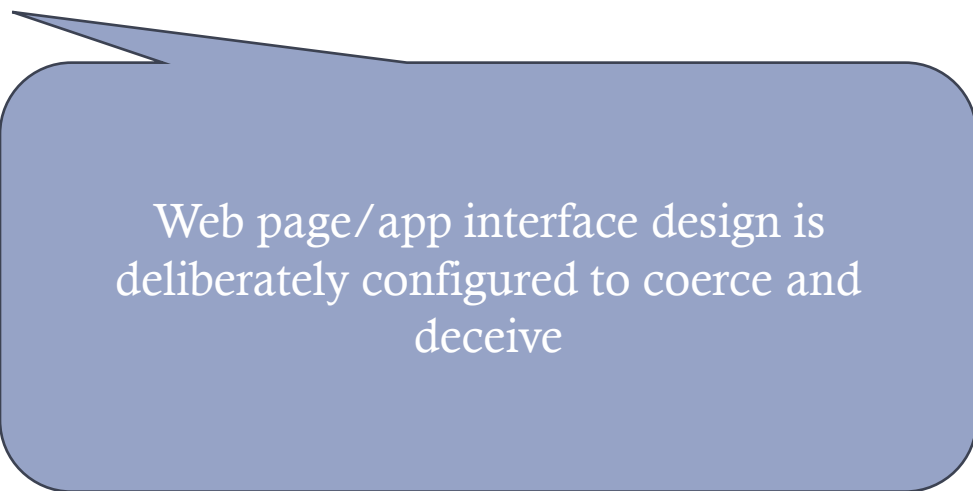
E Ethics

Question:

How would you feel if the company you were working for engaged in legal but questionable practices...?

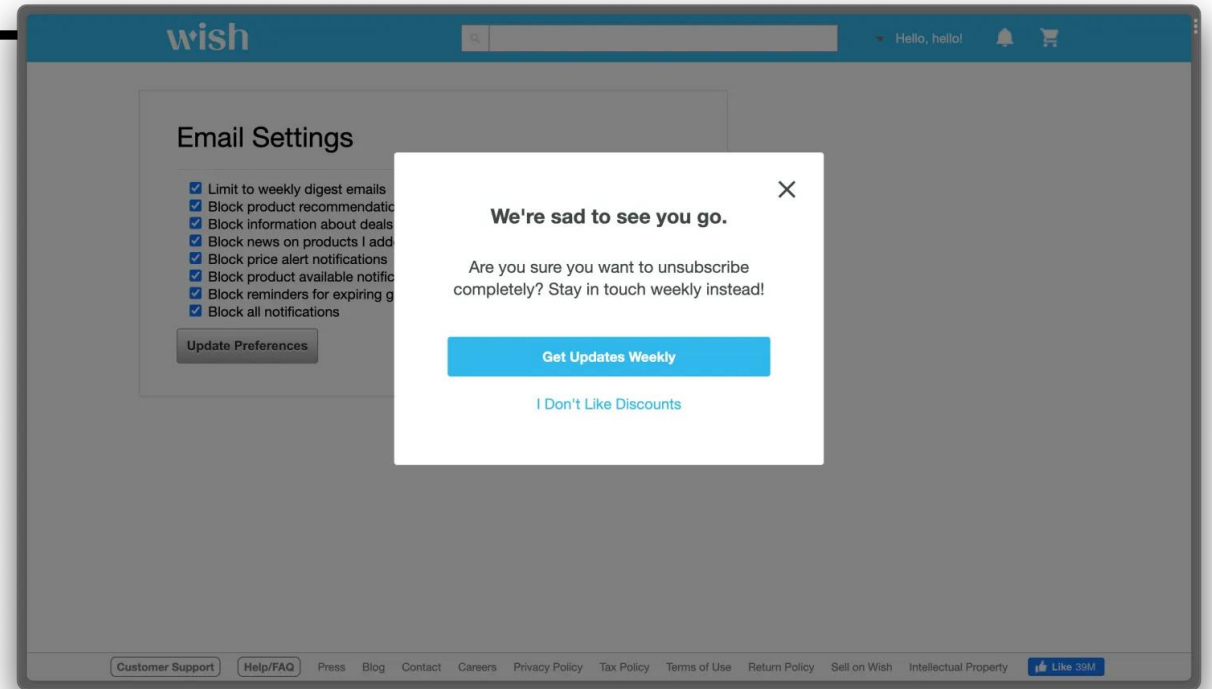
The so-called '**grey area**'

Eg: Deceptive patterns of design



Web page/app interface design is deliberately configured to coerce and deceive

Confirmshaming: Wish.com



Designed to make the user guilty for opting out of notifications.
Definitely a psychological manipulation – manipulating behaviour for their benefit

Just use a simple 'Unsubscribe to Notifications' link...?

Misdirection: London Zoo



The 'Add to basket with donation' button is more prominent and better placed on the page.

This puts subtle pressure on visitors to donate.

	With Donation	Without Donation	
ADULT (AGES 16-64)	£30.50	£27.73	1 ▼
CHILD (3-15 YEARS) ⓘ	£19.85	£18.04	0 ▼
UNDER 3 (AGES 0-2)	£0.00	£0.00	0 ▼
SENIOR (+65) ⓘ	£27.45	£24.95	0 ▼
STUDENT (WITH ID) ⓘ	£27.45	£24.95	0 ▼
DISABLED ADULT (CARER GOES FREE) ⓘ	£27.45	£24.95	0 ▼
DISABLED CHILD (CARER GOES FREE) ⓘ	£19.85	£18.04	0 ▼



Please support our animals

A voluntary donation of 10% will help us to feed and care for our 30,000 animals, many of them **endangered in the wild**. The pandemic has hit us hard, and we need donations to support our amazing animals, from tiny dart frogs to majestic tigers. When you donate, your **whole purchase** is eligible for Gift Aid. This means we can claim an additional 25p for every £1 you spend with us, at **no extra cost to you!**

giftaid it

Add to basket without donation

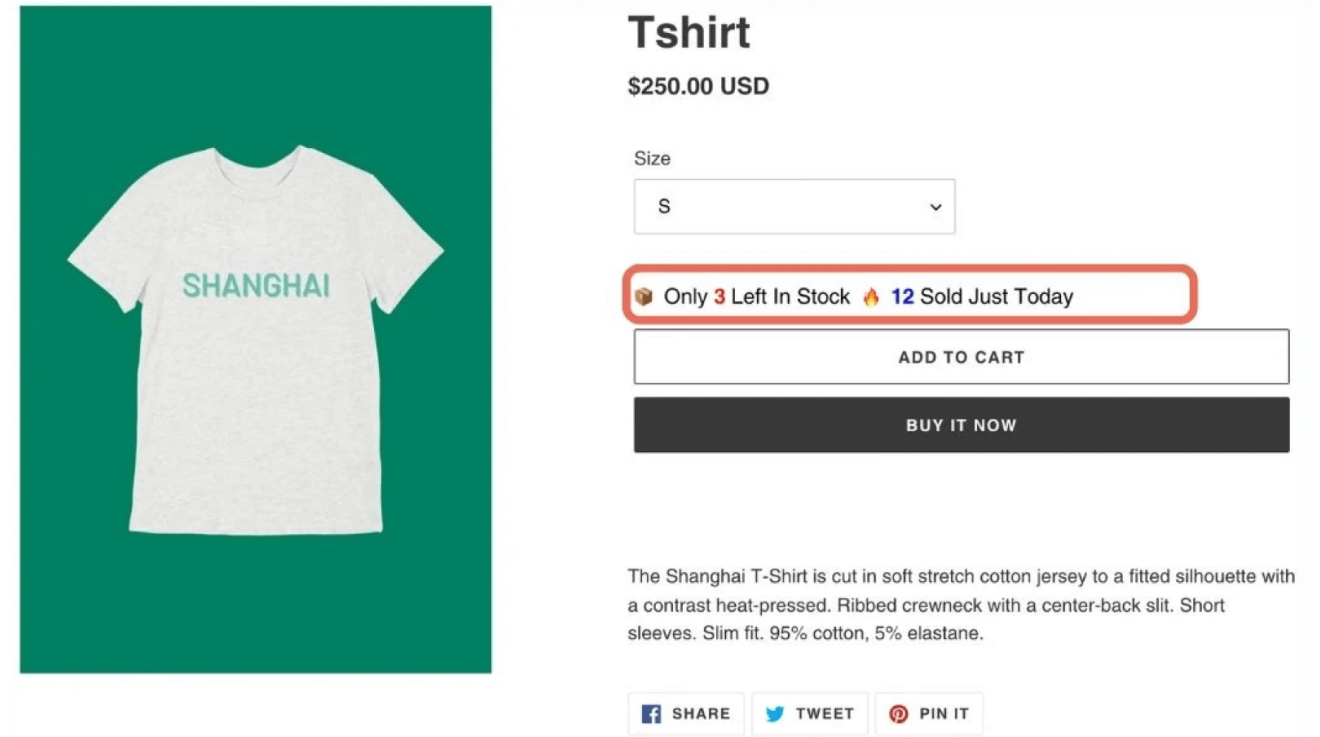
Add to basket with donation

Fake Scarcity: Shopify (+ many others)



Act on fears of users and FOMO

Fake stock message reports. Shop owner can generate fake scarcity and fake sales numbers.



Hidden Subscription: Spotify



Users are signing up for a subscription after the 1 month free.

This fact is hidden in the Ts & Cs

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- 15 hours/month of listening time from our audiobooks subscriber catalogue
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4.3 Trial duration and cancellation.

In the case of any Trial, the corresponding Trial Period shall continue for the period as advertised, subject to section 4.2, above.

Unless cancelled before the end of the Trial Period, or such Trial is in conjunction with a Prepaid Period, you will automatically become a subscriber to the particular Paid Subscription that you chose to sign up to under the Trial and the payment method you provided will automatically be charged the then-current recurring price on the first day following the end of the Trial on a monthly recurring basis or other interval that we disclose to you in advance (any time-capped features of that Paid Subscription will be reduced by the length of the Trial Period). ** IF YOU DO NOT WANT TO PAY THIS CHARGE, YOU MUST CANCEL BEFORE THE END OF THE TRIAL BY A) FOLLOWING THE STEPS DESCRIBED ** [HERE](#) ** IF YOU SIGNED UP FOR THE TRIAL THROUGH SPOTIFY, OR B) IF YOU RECEIVED YOUR TRIAL THROUGH A THIRD PARTY, YOU MUST CANCEL THE APPLICABLE PAID SUBSCRIPTION THROUGH THE THIRD PARTY**.

When
systems/apps/webpages
behave differently to how
users expect....

This is when users feel
tricked.

Can have long term issues
in the loss of users /
customers / players if a
short-term gain of
benefiting the business is
chosen.



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P Professional

The British Computer Society (BCS) Code of Conduct

Association for Computing Machinery (ACM) Code
of Ethics and Professional Conduct

WHAT DOES THIS MEAN FOR YOUR TEAM?

- Questions to ask yourselves....
- How can this product be misused/abused for other purposes?
- How can this project go wrong?

WHAT DOES THIS MEAN FOR YOU AS A PROFESSIONAL?

- Questions...?
- Are you complying with the relevant laws and regulations
- How does your artefact impact society, accessibility, have the potential to be biased
- Are you applying the ethical standards and codes of conduct from professional frameworks (BCS/ACM)
- Are you acting with integrity, professional competence and being accountable

THANK YOU!

- Summary
- We need to be aware of LSEP issues in our roles as computing/games/IT professionals

